TU Clausthal

Requirements Engineering

Exercise – 4

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Deadline: 11.01.2027 – 1:59 PM

Submission location: Moodle Course Page

Submission Guidelines

Please adhere to each guideline. Failure to follow even a single guideline will result in failing the exercise.

1. Please upload two separate valid PDF files
    a. First PDF for the list of Roles, Goals, and BIM (Max 3 pages)
    b. Second PDF containing your AOM Model (Max 1 Page)
2. Make sure that AOM diagram is clearly visible without much effort.
3. Include your full name, program, and matriculation number in the footer of each PDF.
4. Only use 'Verdana' font in size 12 for all text. Use Verdana, size 12, bold fonts for titles.
5. Please review and finalize your submission once you have uploaded all files and then click the Submit button, draft submissions in moodle will not be graded.

Exercise 4 : Agent-Oriented Modeling

Scenario:

You are tasked with developing a bike ride-share system. It allows registered commuters to approach an idle bike and reserve it, after which they use the bike to commute a certain distance (unknown before use). Finally, after the commuter reaches their destination, they end the ride, which triggers automatic computation of the ride fees. This is automatically debited using the commuter's registered payment details.

Task(s):

1. Identify and list the relevant Agents, Roles and Goals (Functional and Quality) of the system. Make sure that you encompass commuter registration, bike reservation/use, and the final payment process, as well as the safety of both (bike and rider).
2. Please use at least one of *time* and *distance* to compute ride cost. The specific computation formula used is up to you.
3. Create an AOM Goal diagram based on your identified Roles and Goals. Please make sure your diagram has at least 3 levels of hierarchy.
4. Create an AOM Behavioral Interface Model that is consistent with your AOM Goal model (For the entire model, not just a subset).

Questions? → etce-re@tu-clausthal.de

We will ignore all emails to our private email accounts!